PES University, Bengaluru

UE24CS341A Software Engineering | Jackfruit Mini-Project

Software Test Plan (STP)

Student Attendance Management System (SAMS)

Project: Student Attendance Management System (SAMS)

Version: 1.2

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Revision History

Version	Date	Author	Change summary
1.0	02-10-2026	Project team	Initial technical baseline
1.2	29-10-2026	Current team	Revised three-member role allocation and review responsibilities.

1. Introduction

2. Test Items

3. Features to be Tested

4. Features Not to be Tested

5. Test Approach / Strategy

5.1 Security Validation

6. Test Environment

7. Test Schedule

8. Test Deliverables

9. Roles and Responsibilities

10. Risks and Mitigation

11. Assumptions & Dependencies

12. Suspension & Resumption Criteria

13. Test Case Management & Traceability

14. Test Metrics & Reporting

15. Approvals

Appendix A: Test Cases

1. Introduction

Purpose: This document defines the test plan for the Student Attendance Management System (SAMS) v1.0. It sets out the objectives, scope, strategy, environment, schedule and responsibilities for testing, and contains the designed unit, integration and system test cases for every use case.

Scope: Testing covers authentication and access control, password reset, attendance marking, the student attendance view, the correction workflow, reports and the defaulter list, course and timetable management, notifications, administration and the audit log, plus the performance, security and accessibility requirements in SRS section 5.

References: SAMS SRS v1.0 (05-09-2026); SAMS Software Architecture and Design Specification v1.0 (30-09-2026); OWASP Testing Guide v4.2; OWASP ASVS 4.0; WCAG 2.1.

Definitions: SRS (Software Requirements Specification), SAD (Software Architecture and Design), RTM (Requirements Traceability Matrix), UT (Unit Test), IT (Integration Test), ST (System Test), UAT (User Acceptance Test), OD (On Duty).

2. Test Items

- AuthModule: login, lockout, logout, password reset, RBAC permission classes
- CourseModule: courses, sections, enrollments, timetable, roll list generation
- AttendanceModule: session marking, duplicate guard, authorised edit, percentage calculation, correction requests
- ReportModule: per-student, per-course, date-range reports, defaulter list, CSV/PDF export
- NotifyModule: in-app notifications, email tasks, nightly low-attendance scan
- AuditModule and Admin: user management, SystemConfig, append-only audit log
- Web UI: Student dashboard, Faculty marking screen, Admin console
- Deployment: Nginx TLS configuration, Docker images, Jenkins pipeline

3. Features to be Tested

Requirement IDs	Feature	Use case in Appendix A
SAMS-F-001 to F-003, SR-003, SR-004	Login, role-based access, lockout, token expiry	UC-01
SAMS-F-004	Password reset through verified email link	UC-02
SAMS-F-005 to F-007	Timetable-driven roll list, marking P/A/OD, no duplicate marking	UC-03
SAMS-F-008	Student views own attendance percentage	UC-04
SAMS-F-009, F-018, SR-005	Correction request and faculty approval	UC-05
SAMS-F-010 to F-013	Reports, defaulter list, CSV/PDF export	UC-06
SAMS-F-014 to F-016	Course, section and timetable management	UC-07
SAMS-F-017, F-018	Low-attendance and pending-request notifications	UC-08
SAMS-F-019 to F-021, NF-004	User and role administration, threshold, audit trail	UC-09
SAMS-NF-001, NF-006	Response time 3 s at p95; 2,000 students and 100 faculty	NFR set
SAMS-NF-003, SR-001, SR-002	Password hashing, TLS	NFR set
SAMS-NF-005	Responsive layout and WCAG 2.1 AA contrast	NFR set

4. Features Not to be Tested

- SAMS-NF-002 (99% uptime over a semester): verified by reviewing uptime monitoring logs after deployment, not by a test case in this cycle.

- Internal behaviour of third-party components: Django, DRF, PostgreSQL, Celery, the SMTP provider (assumed tested by their maintainers; only our integration with them is tested).
- Out-of-scope features listed in SRS 1.2: fees, examinations, biometric hardware, ERP integration.
- Browsers older than the current and previous major versions of Chrome, Firefox, Edge and Safari.

5. Test Approach / Strategy

Levels

- Unit tests (UT): pytest + pytest-django on each module's service functions and models, run on SQLite in memory. Target: 80% line coverage and 70% branch coverage per module, reported by coverage.py in Jenkins.
- Integration tests (IT): DRF APIClient tests across modules (API + service + DB) against PostgreSQL in Docker; Celery in eager mode; Postman/Newman collection for the documented endpoints.
- System tests (ST): end-to-end flows in the browser on the staging deployment, manual first and automated with Selenium for the login, marking and correction flows.
- Acceptance tests (UAT): the course faculty and two student volunteers run the ST scenarios on staging and sign off.

Types

- Functional testing of every SRS requirement, with valid and invalid inputs and boundary values (for example 74.99% vs 75.00% for defaulters, 5th vs 6th failed login).
- Regression testing: the whole UT and IT suite runs on every pull request in Jenkins; merges are blocked on failure.
- Performance testing with Locust against SAMS-NF-001 and NF-006.
- Security testing (section 5.1).
- Usability and accessibility testing with Lighthouse/axe and manual keyboard navigation.

Entry criteria: feature merged to the develop branch with its unit tests; build green in Jenkins; staging environment and seed data available; test cases for the feature reviewed.

Exit criteria: 100% of planned test cases executed; all High-priority requirements passed; 0 open Critical or High defects; coverage targets met; RTM shows every requirement passed or formally waived.

5.1 Security Validation

- Authentication: lockout, password policy, reset token expiry and single use, no user enumeration (UC-01, UC-02).
- Authorisation: server-side RBAC and object-level checks on every endpoint, IDOR attempts with each role's token (ST_SEC_04).
- Transport: TLS 1.2+ only, HTTP to HTTPS redirect, HSTS (ST_SEC_01).
- Secrets and data: passwords hashed with bcrypt, never logged or returned (ST_SEC_02); secrets read from environment.
- Input handling: SQL injection, XSS and malformed JSON fuzzing on all input fields (ST_SEC_05).
- Automated scanning: OWASP ZAP baseline and API scan (ST_SEC_03), SonarQube security hotspots and pip-audit dependency check in every Jenkins build.
- Audit integrity: audit log is append-only and read-only through the API (UC-09).

6. Test Environment

Item	Details
Hardware	Developer laptops (8 GB RAM); staging Linux VM, 2 vCPU, 4 GB RAM
Software	Python 3.12, Django 5, DRF, PostgreSQL 14, Redis 7, Celery 5, Nginx, Docker Compose

Item	Details
Browsers	Chrome, Firefox, Edge (latest), Safari on iOS, Chrome on Android at 360px
Tools	pytest, pytest-django, coverage.py, factory_boy, freezegun, Postman/Newman, Selenium, Locust, OWASP ZAP, testssl.sh, Lighthouse, SonarQube, Jenkins, Jira (defects)
Test data	Seed script creating 1 admin, 100 faculty, 2,000 students, 20 courses, sections and a 6-week timetable with generated attendance; MailHog to capture email

7. Test Schedule

Milestone	Date
Test plan and test case design	02-10-2026
Test environment setup (Docker staging, seed data, MailHog)	01-10-2026 to 07-10-2026
Sprint 1 testing: Auth, Course and Timetable (UC-01, UC-02, UC-07)	01-10-2026 to 14-10-2026
Sprint 2 testing: Marking, Student view, Corrections (UC-03, UC-04, UC-05)	15-10-2026 to 28-10-2026
Sprint 3 testing: Reports, Notifications, Admin and Audit (UC-06, UC-08, UC-09)	29-10-2026 to 11-11-2026
Performance and security test cycle	05-11-2026 to 11-11-2026
UAT	12-11-2026 to 16-11-2026
Test summary report	17-11-2026

8. Test Deliverables

- Test Plan (this document)
- Test cases (Appendix A) and automated test scripts in the GitHub repository (tests/ folder in each module)
- Test data seed scripts
- Test execution logs and Jenkins coverage reports
- Defect reports in Jira (issue type Bug, linked to the story)
- Test Summary Report and updated RTM

9. Roles and Responsibilities

Role	Name	Responsibility
Developer	Kundhan Vudutha	Implement features/fixes and automated regression checks; provide reproducible builds and explain code changes.
Tester	Tarun J	Execute UC-01 through UC-09, integration/system/security/performance/accessibility cases; record actual results and retest defects.
QA	Anish	Own test-plan review, traceability, coverage/evidence checks, defect quality and release-readiness review.
Product Owner	Prof. Rajeshwari Adrakatti	Review results and provide actual UAT sign-off.

10. Risks and Mitigation

Risk	Mitigation
Features finish late in the sprint, leaving little time to test	Unit tests written with each feature; definition of done in Jira requires tests

Risk	Mitigation
Staging environment unavailable	Docker Compose lets the same stack run on any team laptop
Developers testing their own code miss their own blind spots	Cross-testing: each member executes the system tests for another member's module
Email delivery hard to verify	MailHog captures all outgoing mail on staging
Realistic load hard to generate	Locust scripts with seeded data at full target scale

11. Assumptions & Dependencies

- Faculty and student volunteers are available for UAT in the scheduled week.
- A free-tier or lab VM is available for staging with a TLS certificate (self-signed acceptable for staging).
- The default minimum-attendance threshold is 75% unless the department states otherwise.
- SRS v1.0 and SAD v1.0 remain the baseline; changes go through the Jira backlog and update this plan.

12. Suspension & Resumption Criteria

Suspend testing if: the staging environment is unavailable for more than 4 hours; the build is unstable and blocks more than 30% of test cases; a Critical security defect is found (stop and fix before continuing).

Resume testing when: blocking defects are fixed and verified; the environment is stable; a smoke run of UC-01 and UC-03 passes.

13. Test Case Management & Traceability

Test cases live in Appendix A and as automated tests in the repository; each automated test name contains its test case ID. Defects are logged in Jira and linked to the failing test case and the story. The table below extends the SRS RTM with the designed test cases.

Requirement	SRS test ref	Designed test cases
SAMS-F-001	TC-Auth-01	UT_AUTH_01, UT_AUTH_05, IT_AUTH_01, ST_AUTH_01
SAMS-F-002	TC-Auth-02	IT_AUTH_02, IT_ADM_03
SAMS-F-003	TC-Auth-03	UT_AUTH_02, UT_AUTH_03, UT_AUTH_04
SAMS-F-004	TC-Auth-04	UT_PWR_01 to UT_PWR_05, IT_PWR_01 to IT_PWR_03, ST_PWR_01
SAMS-F-005, F-016	TC-Att-01	UT_ATT_01, UT_ATT_02, ST_CRS_01
SAMS-F-006	TC-Att-02	UT_ATT_03, UT_ATT_05, UT_ATT_06, IT_ATT_01, IT_ATT_02, ST_ATT_01
SAMS-F-007	TC-Att-03	UT_ATT_04, UT_ATT_07
SAMS-F-008	TC-Att-04	UT_VIEW_01 to UT_VIEW_04, IT_VIEW_01 to IT_VIEW_03, ST_VIEW_01
SAMS-F-009	TC-Att-05	UT_COR_01 to UT_COR_06, IT_COR_01, IT_COR_02, ST_COR_01
SAMS-F-010 to F-013	TC-Rep-01	UT_REP_01 to UT_REP_06, IT_REP_01, IT_REP_02, ST_REP_01
SAMS-F-014 to F-016	-	UT_CRS_01 to UT_CRS_05, IT_CRS_01 to IT_CRS_03, ST_CRS_01
SAMS-F-017, F-018	-	UT_NOT_01 to UT_NOT_04, IT_NOT_01 to IT_NOT_03, ST_NOT_01
SAMS-F-019 to F-021	-	UT_ADM_01 to UT_ADM_04, IT_ADM_01 to IT_ADM_03, ST_ADM_01
SAMS-NF-001, NF-006	TC-Perf-01, TC-Perf-02	ST_PERF_01, ST_PERF_02
SAMS-NF-003, SR-001, SR-002	TC-Sec-01 to TC-Sec-03	ST_SEC_01, ST_SEC_02, UT_ADM_01

Requirement	SRS test ref	Designed test cases
SAMS-SR-003	TC-Sec-04	IT_AUTH_03, IT_AUTH_04
SAMS-SR-004	TC-Sec-05	IT_AUTH_02, ST_SEC_04
SAMS-SR-005, NF-004	TC-Sec-06, TC-OPS-01	UT_ADM_03, UT_ADM_04, IT_ADM_01, ST_OPS_01
SAMS-NF-005	TC-UX-01	ST_VIEW_01, ST_UX_01

14. Test Metrics & Reporting

Planned test cases: 89 in total (46 unit, 25 integration, 18 system).

- Percentage of test cases executed and passed/failed, per use case
- Unit test line and branch coverage per module (Jenkins coverage report)
- Defect count by severity, defect density per module, defect age
- Requirement coverage from the RTM

Reporting: status updated at the daily stand-up and in Jira; a Test Summary Report at the end of the cycle.

15. Approvals

Role	Name	Signature / Date
Developer	Kundhan Vudutha	
Tester	Tarun J	
QA	Anish	
Product Owner	Prof. Rajeshwari Adrakatti	

Appendix A: Test Cases

ID convention: UT = unit test, IT = integration test, ST = system test, followed by the module code and number (for example UT_AUTH_01). Actual Result and Test Result are filled in during execution; cases are listed here as designed and not yet executed.

UC-01 Login and Access Control (SAMS-F-001, F-002, F-003, SR-003, SR-004)

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT_AUTH_01	AuthModule / authenticate()	Valid credentials return a user and tokens (SAMS-F-001)	Test DB with seeded student user	1: Call authenticate(institute_id, password) 2: Inspect returned user and tokens	ID: PES1UG24AM999, Password: Test@1234	Returns user object with role STUDENT; access and refresh JWTs issued; failed_attempts = 0	Not yet executed	Pending
UT_AUTH_02	AuthModule / authenticate()	Wrong password is rejected and counter increments	Seeded user, failed_attempts = 0	1: Call authenticate() with wrong password 2: Read user.failed_attempts	ID: PES1UG24AM999, Password: wrong123	Raises InvalidCredentials; failed_attempts = 1; no token issued	Not yet executed	Pending
UT_AUTH_03	AuthModule / lockout	Account locks after 5 consecutive failures (SAMS-F-003)	Seeded user, failed_attempts = 4	1: Call authenticate() with wrong password 2: Call authenticate() with correct password	Password: wrong123, then Test@1234	Second call raises AccountLocked; locked_until = now + 15 min; lockout event logged	Not yet executed	Pending
UT_AUTH_04	AuthModule / lockout	Lock expires after 15 minutes	User locked_until = now - 1 s (time frozen with freezegun)	1: Call authenticate() with correct password	Password: Test@1234	Login succeeds; failed_attempts reset to 0	Not yet executed	Pending
UT_AUTH_05	AuthModule / validation	Empty fields are rejected by serializer	None	1: Validate LoginSerializer with empty password	ID: PES1UG24AM999, Password: (empty)	Serializer invalid; error "This field may not be blank."	Not yet executed	Pending
IT_AUTH_01	API Router + AuthModule	Login endpoint returns role and tokens	API server running with test DB	1: POST /api/auth/login/ with valid JSON 2: Check status and body	{"institute_id":"FAC001", "password":"Test@1234 "}	HTTP 200; body has access, refresh, role = FACULTY	Not yet executed	Pending
IT_AUTH_02	API Router / RBAC	Student token cannot call faculty endpoint (SAMS-F-002, SR-004)	Student logged in, holds access token	1: POST /api/sessions/412/attendance/ with student token	Bearer <student JWT>, any records payload	HTTP 403 {"error": {"code":"NOT_FACULTY"}}; no records written	Not yet executed	Pending
IT_AUTH_03	API Router / tokens	Expired token is rejected (SAMS-SR-003)	Access token with lifetime set to 1 s in test settings	1: Login 2: Wait 2 s 3: GET /api/students/me/attendance/	Expired access JWT	HTTP 401 token_not_valid	Not yet executed	Pending
IT_AUTH_04	API Router / logout	Logout blacklists refresh token	User logged in	1: POST /api/auth/logout/ with refresh token 2: POST /api/auth/refresh/ with same token	Refresh JWT from login	Step 1: HTTP 205. Step 2: HTTP 401 token is blacklisted	Not yet executed	Pending
ST_AUTH_01	Login page (UI)	End-to-end login lands on the correct dashboard	Chrome browser; seeded Student, Faculty, Admin accounts	1: Open https://sams.local/login 2: Enter ID and password 3: Click Sign in 4: Repeat for each role	STU001 / FAC001 / ADM001, Password: Test@1234	Each user sees their own role dashboard with "Welcome, <name>"; invalid password shows "Invalid ID or password"	Not yet executed	Pending

UC-02 Password Reset (SAMS-F-004)

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT_PWR_01	AuthModule / reset token	Reset token is generated and stored hashed	Seeded user with email	1: Call request_reset(email) 2: Inspect DB token row	kundhan.test@pes.edu	One token row with hashed value and expiry = now + 30 min; email task queued	Not yet executed	Pending
UT_PWR_02	AuthModule / reset token	Expired token is rejected	Token created 31 min ago (frozen time)	1: Call confirm_reset(token, new_password)	New password: NewPass@123	Raises TokenExpired; password unchanged	Not yet executed	Pending
UT_PWR_03	AuthModule / reset token	Token can only be used once	Valid unused token	1: confirm_reset() once 2: confirm_reset() again with same token	New password: NewPass@123	First call succeeds; second raises TokenUsed	Not yet executed	Pending
UT_PWR_04	AuthModule / password policy	Weak new password is rejected	Valid token	1: confirm_reset(token, "12345")	New password: 12345	ValidationError listing password rules; token not consumed	Not yet executed	Pending
UT_PWR_05	AuthModule / enumeration	Unknown email gets same response as known email	None	1: request_reset() with unknown email 2: Compare result with known email	nobody@pes.edu	Same success result; no email queued; no token row created	Not yet executed	Pending
IT_PWR_01	API + NotifyModule	Reset request queues an email with a one-time link	Celery in eager mode, locmem email backend	1: POST /api/auth/password-reset/ 2: Inspect outbox	{"email":"kundhan.test@pes.edu"}	HTTP 202; one email with link containing token	Not yet executed	Pending
IT_PWR_02	API / confirm	Confirm endpoint sets new password	Valid token from IT_PWR_01	1: POST /api/auth/password-reset/confirm/ 2: Login with new password	{"token":"<t>","new_password":"NewPass@123"}	HTTP 200; login with new password returns 200; old password returns 401	Not yet executed	Pending
IT_PWR_03	API / confirm	Reused token returns 410	Token already used	1: POST confirm again	Same token	HTTP 410 TOKEN_EXPIRED_OR_USED	Not yet executed	Pending
ST_PWR_01	Forgot password (UI)	User resets password from login page	Browser; test mailbox (MailHog)	1: Click "Forgot password" 2: Enter email, submit 3: Open link from MailHog 4: Set new password 5: Log in	Email: kundhan.test@pes.edu, New: NewPass@123	Message "If the account exists, a link has been sent"; reset page accepts new password; login succeeds	Not yet executed	Pending

UC-03 Mark Attendance (SAMS-F-005, F-006, F-007)

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT_ATT_01	CourseModule / slots_for()	Returns only the faculty's sessions for today (SAMS-F-005)	Faculty with 2 slots on Monday, 1 on Tuesday; date = Monday	1: Call slots_for(faculty, monday)	Faculty FAC001, date 2026-10-05	Exactly 2 sessions returned, ordered by start time	Not yet executed	Pending
UT_ATT_02	CourseModule / roster()	Roster contains only enrolled students	Section F with 60 students; 5 students in section E	1: Call roster(section F)	Section F	60 students returned; none from section E	Not yet executed	Pending
UT_ATT_03	AttendanceModule / submit()	Saves P, A and OD statuses (SAMS-F-006)	Unsubmitted session with 3 students	1: Call submit(session, records) 2: Query records	S1 = P, S2 = A, S3 = OD	3 records saved with correct statuses; session.submitted =	Not yet executed	Pending

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
						True		
UT_ATT_04	AttendanceModule / submit()	Second submit is blocked (SAMS-F-007)	Session already submitted	1: Call submit() again	Same records	Raises AlreadySubmitted; record count unchanged	Not yet executed	Pending
UT_ATT_05	AttendanceModule / validation	Invalid status is rejected	Unsubmitted session	1: Call submit() with status "X"	S1 = X	Raises ValidationError INVALID_STATUS; nothing saved (transaction rolled back)	Not yet executed	Pending
UT_ATT_06	AttendanceModule / validation	Student outside roster is rejected	Unsubmitted session for section F	1: Call submit() including a section E student	Student from section E = P	Raises ValidationError NOT_IN_ROSTER; nothing saved	Not yet executed	Pending
UT_ATT_07	AttendanceModule / edit()	Authorised edit changes status and writes audit entry	Submitted session; record S2 = A	1: Call edit(record, "P", editor=faculty, reason)	Reason: "Late entry verified"	Record status = P; one AuditLog row with old = A, new = P, editor = faculty	Not yet executed	Pending
IT_ATT_01	API + AttendanceModule	POST attendance returns summary counts	Faculty token; session 412 with 60 students	1: POST /api/sessions/412/attendance/	54 P, 5 A, 1 OD	HTTP 201 {"saved":60,"present":54,"absent":5,"on_duty":1}	Not yet executed	Pending
IT_ATT_02	API + AttendanceModule	Faculty not assigned to the session gets 403	Token for FAC002 (teaches another section)	1: POST /api/sessions/412/attendance/	Valid payload	HTTP 403 NOT_SESSION_FACULTY	Not yet executed	Pending
ST_ATT_01	Faculty marking screen (UI)	Full roll call of 60 students saved quickly (SAMS-F-006)	Faculty logged in; session scheduled now	1: Open Today's sessions 2: Open session 3: Mark 6 students Absent, 1 On Duty 4: Click Submit 5: Time steps 2 to 4	60-student section F	Success toast with counts; whole flow under 60 s; server save under 2 s (network tab)	Not yet executed	Pending

UC-04 View Own Attendance (SAMS-F-008)

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT_VIEW_01	AttendanceModule / percentage()	Percentage counts P and OD as attended	10 sessions: 7 P, 1 OD, 2 A	1: Call percentage(student, course)	As described	Returns 80.00	Not yet executed	Pending
UT_VIEW_02	AttendanceModule / percentage()	No sessions held returns 0 without error	Course with no sessions yet	1: Call percentage()	0 sessions	Returns 0.0 and held = 0; no ZeroDivisionError	Not yet executed	Pending
UT_VIEW_03	AttendanceModule / percentage()	Rounding to two decimals	3 sessions: 2 P, 1 A	1: Call percentage()	As described	Returns 66.67	Not yet executed	Pending
UT_VIEW_04	AttendanceModule / threshold flag	below_threshold flag uses SystemConfig	Threshold = 75; student at 74.5%	1: Call summary(student)	Threshold 75	below_threshold = True	Not yet executed	Pending
IT_VIEW_01	API / students/me	Student sees only own data	Two students with records	1: GET /api/students/me/attendance/ as student A	Student A token	HTTP 200; list contains only student A's courses	Not yet executed	Pending

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
IT_VIEW_02	API / object permission	Student cannot read another student by ID	Student A token	1: GET /api/students/{B_id}/attendance/	B_id = id of student B	HTTP 403 (or 404); no data of B returned	Not yet executed	Pending
IT_VIEW_03	API + AttendanceModule	New submission reflected immediately	Student at 7/10	1: Faculty submits session 11 with student P 2: Student GETs attendance	Session 11 = P	held = 11, attended = 8, percentage = 72.73 (well inside 5-minute target)	Not yet executed	Pending
ST_VIEW_01	Student dashboard (UI)	Dashboard cards show per-course percentage with text label	Student with 3 courses, one below 75%	1: Log in as student 2: View dashboard on desktop 3: Resize to 360px width	STU001	Three cards; low course labelled "Below threshold" in text and colour; layout readable at 360px	Not yet executed	Pending

UC-05 Attendance Correction Request (SAMS-F-009, F-018, SR-005)

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT_COR_01	AttendanceModule / create_request()	Student creates request on own Absent record	Record S1 = A in submitted session	1: Call create_request(student, record, "OD", reason)	Reason: "Hackathon, letter attached"	Request saved with state PENDING; faculty notification created	Not yet executed	Pending
UT_COR_02	AttendanceModule / create_request()	Empty reason is rejected	Record S1 = A	1: Call create_request() with reason ""	Reason: (empty)	ValidationError REASON_REQUIRED	Not yet executed	Pending
UT_COR_03	AttendanceModule / create_request()	Cannot open a second request on the same record	Pending request exists for record	1: Call create_request() again	Same record	Raises OpenRequestExists	Not yet executed	Pending
UT_COR_04	AttendanceModule / decide()	Approve updates record and audit log	Pending request A -> OD	1: Call decide(request, faculty, APPROVE)	Decision APPROVE	Record status = OD; request APPROVED; AuditLog row old = A, new = OD; student notified	Not yet executed	Pending
UT_COR_05	AttendanceModule / decide()	Reject keeps record unchanged	Pending request	1: Call decide(request, faculty, REJECT, note)	Note: "No proof provided"	Record unchanged; request REJECTED with note; student notified	Not yet executed	Pending
UT_COR_06	AttendanceModule / decide()	Decided request cannot be decided again	Request already APPROVED	1: Call decide(..., REJECT)	Decision REJECT	Raises AlreadyDecided; record unchanged	Not yet executed	Pending
IT_COR_01	API / corrections	Student cannot raise request on someone else's record	Student A token, record belongs to B	1: POST /api/corrections/	{"record_id":<B record>,"requested_status":"P","reason":"x"}	HTTP 403; no request created	Not yet executed	Pending
IT_COR_02	API / corrections	Only section faculty can decide	Pending request in section F; token of FAC002 (section E)	1: PATCH /api/corrections/77/	{"decision":"APPROVE"}	HTTP 403; request still PENDING	Not yet executed	Pending
ST_COR_01	Correction workflow (UI)	Full request and approval flow across two users	Student and faculty accounts; Absent record exists	1: Student opens course, clicks "Request correction" on the absent date 2: Enters reason, submits 3: Faculty opens Pending requests,	Reason: "Hackathon, letter attached"	Faculty sees request with reason; after approval student sees status OD and updated percentage; notification shown to both	Not yet executed	Pending

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
				approves 4: Student refreshes dashboard				

UC-06 Reports and Defaulter List (SAMS-F-010 to F-013)

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT_REP_01	ReportModule / defaulters()	Lists students strictly below threshold	Students at 74.99%, 75.00%, 80%	1: Call defaulters(course, threshold=75)	Threshold 75	Only the 74.99% student returned	Not yet executed	Pending
UT_REP_02	ReportModule / defaulters()	Threshold outside 0 to 100 rejected	None	1: Call defaulters(course, threshold=120)	Threshold 120	ValidationError THRESHOLD_RANGE	Not yet executed	Pending
UT_REP_03	ReportModule / course_report()	Course report aggregates per student	Course with 5 students and 10 sessions	1: Call course_report(course)	Seeded data	5 rows; held = 10 for each; attended matches seeded data	Not yet executed	Pending
UT_REP_04	ReportModule / date range	Date-range report includes both end dates	Sessions on 1, 5 and 10 Oct	1: Call range_report(from=1 Oct, to=5 Oct)	From 2026-10-01 to 2026-10-05	Sessions on 1 and 5 Oct counted; 10 Oct excluded	Not yet executed	Pending
UT_REP_05	ReportModule / export	CSV export has header and correct rows	Defaulter list of 3 students	1: Call export(rows, "csv") 2: Parse CSV	3 rows	Header srn,name,course,held,attended,percentage plus 3 data rows	Not yet executed	Pending
UT_REP_06	ReportModule / export	PDF export produces a valid PDF	Same 3 rows	1: Call export(rows, "pdf")	3 rows	Output starts with %PDF; contains course code and 3 SRNs	Not yet executed	Pending
IT_REP_01	API / reports	Faculty can only report on own sections	FAC001 token (section F)	1: GET /api/reports/defaulters/?section=E	Section E	HTTP 403	Not yet executed	Pending
IT_REP_02	API / reports	CSV download headers correct	Admin token	1: GET /api/reports/defaulters/?course=UE24CS341A&format=csv	format=csv	HTTP 200; Content-Type text/csv; Content-Disposition attachment	Not yet executed	Pending
ST_REP_01	Reports page (UI)	Admin generates and downloads defaulter list	Admin logged in; seeded semester data	1: Open Reports 2: Choose course, threshold 75 3: Click Generate 4: Click Download PDF	UE24CS341A, 75	Table shows defaulters sorted by percentage; PDF downloads and matches table	Not yet executed	Pending

UC-07 Course and Timetable Management (SAMS-F-014 to F-016)

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT_CRS_01	CourseModule / Course	Course code must be unique	Course UE24CS341A exists	1: Create another course with same code	Code UE24CS341A	IntegrityError / validation error "course code already exists"	Not yet executed	Pending

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT_CRS_02	CourseModule / TimetableSlot	End time must be after start time	Section F exists	1: Create slot 10:00 to 09:00	Mon 10:00 to 09:00	ValidationError END_BEFORE_START	Not yet executed	Pending
UT_CRS_03	CourseModule / clash check	Faculty cannot be double-booked	FAC001 has Mon 09:00 to 10:00	1: Create slot for FAC001 Mon 09:30 to 10:30	As described	ValidationError FACULTY_CLASH	Not yet executed	Pending
UT_CRS_04	CourseModule / clash check	Section cannot have two classes at once	Section F has Mon 09:00 to 10:00	1: Create another slot for section F Mon 09:00	As described	ValidationError SECTION_CLASH	Not yet executed	Pending
UT_CRS_05	CourseModule / Enrollment	Student enrolled once per section	Student enrolled in F	1: Enroll same student in F again	STU001, section F	Validation error; one enrollment row	Not yet executed	Pending
IT_CRS_01	API / admin courses	Admin creates course, section and slot	Admin token	1: POST /api/courses/ 2: POST /api/sections/ 3: POST /api/timetable/	UE24CS341A, F, Mon 09:00 to 10:00, FAC001	All HTTP 201; FAC001 sees the slot in /api/sessions/today/ on Monday	Not yet executed	Pending
IT_CRS_02	API / RBAC	Faculty cannot create courses	Faculty token	1: POST /api/courses/	Any course	HTTP 403	Not yet executed	Pending
IT_CRS_03	API / CSV import	Bulk enrollment from CSV	Admin token; CSV of 60 SRNs, 2 invalid	1: POST /api/sections/F/enroll-csv/	enroll.csv	HTTP 200 {"enrolled":58,"errors":[2 rows with reason]}	Not yet executed	Pending
ST_CRS_01	Admin console (UI)	Admin sets up a new section timetable end to end	Admin logged in	1: Create course 2: Create section 3: Import students CSV 4: Add 3 weekly slots 5: Log in as faculty on a slot day	Course UE24CS341A, section F	Faculty sees the session with the imported 60-student roster	Not yet executed	Pending

UC-08 Notifications (SAMS-F-017, F-018)

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT_NOT_01	NotifyModule / nightly_scan()	Creates notifications for students below threshold	2 students below 75%, 3 above	1: Run nightly_scan()	Threshold 75	2 Notification rows; 2 emails queued	Not yet executed	Pending
UT_NOT_02	NotifyModule / nightly_scan()	Does not re-notify the same student the same day	Scan already ran today	1: Run nightly_scan() again	Same data	No new notifications	Not yet executed	Pending
UT_NOT_03	NotifyModule / notify()	In-app notification saved even if email fails	SMTP backend raises error	1: Call notify(user, msg)	Any message	Notification row exists; email task scheduled for retry	Not yet executed	Pending
UT_NOT_04	NotifyModule / email task	Retries 3 times then dead-letters	SMTP always fails	1: Run email task	Any	3 retries with backoff; dead-letter row created	Not yet executed	Pending
IT_NOT_01	AttendanceModule + NotifyModule	Correction request notifies section faculty	Celery eager; locmem email	1: Student creates correction request	Any valid request	Faculty has 1 unread notification and 1 email in outbox	Not yet executed	Pending
IT_NOT_02	API / notifications	User marks notification as	User with 1 unread	1: GET /api/notifications/	-	Step 2 HTTP 200; unread count =	Not yet	Pending

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
		read	notification	2: PATCH /api/notifications/{id}/ {"read":true}		0	executed	
IT_NOT_03	API / notifications	User cannot read another user's notifications	Notification belongs to user B	1: GET /api/notifications/{B_id}/ as user A	-	HTTP 404	Not yet executed	Pending
ST_NOT_01	Notification bell (UI)	Low-attendance alert visible to student	Student below threshold; nightly job run	1: Log in as student 2: Click bell icon	STU002	Alert "Your attendance in UE24CS341A is 70%, below 75%"; email received in MailHog	Not yet executed	Pending

UC-09 Administration and Audit (SAMS-F-019 to F-021, SR-005, NF-004)

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT_ADM_01	Admin / users	Admin creates user with exactly one role	None	1: Call create_user(id, role=FACULTY)	FAC010	User saved with role FACULTY; password stored as bcrypt hash	Not yet executed	Pending
UT_ADM_02	Admin / SystemConfig	Threshold must be between 0 and 100	None	1: Set min_attendance_pct = 150	150	ValidationError	Not yet executed	Pending
UT_ADM_03	AuditModule / AuditLog	Audit rows cannot be updated or deleted through the model API	One audit row	1: Call row.save() after change 2: Call row.delete()	-	Both raise PermissionDenied / ImmutableRecord	Not yet executed	Pending
UT_ADM_04	AuditModule / log()	Every attendance edit creates exactly one audit row	Submitted record	1: Call edit() twice	A -> P -> OD	2 audit rows in order, with editor and timestamps	Not yet executed	Pending
IT_ADM_01	API / audit	Audit endpoint is read-only	Admin token	1: PUT /api/audit/1/ 2: DELETE /api/audit/1/	-	HTTP 405 for both; row unchanged	Not yet executed	Pending
IT_ADM_02	API / audit	Non-admin cannot read audit log	Faculty token	1: GET /api/audit/	-	HTTP 403	Not yet executed	Pending
IT_ADM_03	API / users	Role change by faculty through profile is ignored	Faculty token	1: PATCH /api/users/me/ {"role":"ADMIN"}	{"role":"ADMIN"}	HTTP 200 or 400 but role remains FACULTY	Not yet executed	Pending
ST_ADM_01	Admin console (UI)	Admin traces an edited attendance mark	One record edited by faculty	1: Admin opens Audit log 2: Filter by student and date	STU001, 2026-10-05	Row shows editor FAC001, old A, new P, timestamp	Not yet executed	Pending

Non-functional and Security Tests (SAMS-NF-001 to NF-006, SR-001 to SR-005)

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
ST_PERF_01	Whole system	Page load p95 within 3 s	Staging on Docker; 2,000	1: Run Locust with 100 concurrent	100 users, 10 min	p95 response time 3 s or less;	Not yet	Pending

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
	(Locust)	under normal load (SAMS-NF-001)	students, 100 faculty seeded	users for 10 min on dashboard and marking flows		error rate below 1%	executed	
ST_PERF_02	Whole system (Locust)	Scale target without architecture change (SAMS-NF-006)	Same as above	1: Ramp to 300 concurrent users	300 users	No 5xx errors; p95 under 3 s or documented bottleneck	Not yet executed	Pending
ST_SEC_01	Nginx / TLS	Plain HTTP is redirected and TLS 1.2+ enforced (SAMS-SR-001)	Staging with certificate	1: curl http://host 2: Run testssl.sh / sslscan	-	HTTP 301 to HTTPS; TLS 1.0/1.1 disabled; certificate valid	Not yet executed	Pending
ST_SEC_02	AuthModule / logs	Passwords never appear in logs or responses (SAMS-SR-002)	Debug log level on staging	1: Perform logins and resets 2: grep logs and API responses for test password	Password: Test@1234	No match found	Not yet executed	Pending
ST_SEC_03	API (OWASP ZAP)	Automated scan of API for common vulnerabilities	Staging; ZAP with authenticated context	1: Run ZAP baseline and API scan with OpenAPI spec	openapi.json	No High findings; Medium findings triaged in Jira	Not yet executed	Pending
ST_SEC_04	API (manual pen test)	Broken access control attempts (SAMS-SR-004)	Tokens for each role	1: Replay every Admin and Faculty endpoint with Student token 2: Change IDs in URLs (IDOR)	Endpoint list from OpenAPI	All return 403/404; no cross-user data	Not yet executed	Pending
ST_SEC_05	Login (fuzzing)	Injection and malformed input on login and forms	Staging	1: Send SQL injection and XSS payloads to login, reason and search fields	' OR 1=1 --, <script>alert(1)</script>	No SQL error, no script execution; inputs escaped; HTTP 400 for malformed	Not yet executed	Pending
ST_UX_01	UI (axe / Lighthouse)	Accessibility and responsive check (SAMS-NF-005)	Core screens available	1: Run Lighthouse accessibility audit on 4 core screens 2: Check at 360px width	-	Accessibility score 90 or above; no contrast failures; no horizontal scroll at 360px	Not yet executed	Pending
ST_OPS_01	AuditModule / retention	Audit history retained for academic year (SAMS-NF-004)	Audit rows dated across the year	1: Query audit log for earliest date	-	Rows from start of academic year present; no purge job deletes them	Not yet executed	Pending